Do Agent Skills Teach Accessibility? — Research Plan

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A 5-page short paper for SE4All 2027 at ICSE, due 13 November 2026 (AoE).

Research questions

1. How often do UI skills mention accessibility?
2. How specific is the guidance, and does any of it conflict with accessibility?
3. Do popular frontend skills change the accessibility of generated UIs?

Method

1. **Corpus:** pull UI skills from GitSkills and ClawHub, deduplicate them, then filter with keywords plus an LLM check validated on about 200 hand-labelled skills.
2. **Mining:** run a lexicon pass (WCAG, ARIA, contrast, keyboard), then manually code about 150 skills as none, generic, specific, or verification, and flag conflicting advice. Two annotators, Cohen's kappa.
3. **Experiment:** compare no skill, the top 5–8 frontend skills, and an accessibility skill, using 10 test cases, 2–3 models, and 3 samples each.
4. **Evaluation:** use Microsoft's a11y-llm-eval harness to measure WCAG pass rate and failures per page.

Related work

- Microsoft A11y LLM Eval: the baseline harness, which shows an accessibility skill improves results.
- From Anatomy to Smells (arXiv 2607.01456): an empirical study of SKILL.md quality.
- Human or LLM? (arXiv 2503.15885): accessibility of LLM-generated web code.

Timeline

1. Week 1: corpus and experiment runs.
2. Week 2: manual coding.
3. Week 3: writing.